

Bradley Pfeil

Adelaide, SA ✦ +61 421 080 007 ✦ Brad180795@gmail.com
Personal Site ✦ LinkedIn ✦ GitHub

Education

The University of Adelaide

Bachelor of Computer Science (Advanced)

2022 – 2024

Work Experience

Data Scientist – Streamline (Aurizn / Lunio)

2025 – Current

Lead Data Scientist owning end-to-end design of a revenue management platform that identifies pricing improvement opportunities through mathematical modelling and optimisation. Built a two-stage statistical model (hurdle classifier + gradient-boosted regressor with uncertainty quantification) to estimate demand transitions, and a Markov Decision Process solver that computes optimal pricing policies, revenue forecasts, and scenario analysis via forward simulation. Applied econometric and causal inference techniques to estimate heterogeneous price elasticities, and designed a counterfactual data augmentation pipeline to improve model generalisation beyond observed price variation. Translated these capabilities into production software deployed on AWS (Docker, ECS, SQS, Lambda), with CI/CD, automated testing, model validation, and code quality governance. Engaged directly with clients and stakeholders to explain model outputs, define domain-specific reward functions, and align pricing recommendations with business objectives. Identified and resolved scaling bottlenecks, halving inference time, and implemented a cost monitoring framework that reduced monthly AWS spend by 20%+. Founded and lead a journal review club, fostering team growth through discussion of AI/ML research.

Intern / Graduate Data Scientist – Aurizn

2023 – 2025

Developed LLM-based recommender systems with SHAP-driven explainability to surface model decision logic for non-technical stakeholders. Engineered data ingestion and automated retraining pipelines for a dynamic pricing system, ensuring consistent model accuracy across changing data distributions while reducing manual maintenance overhead.

Projects

Adaptive Database Optimiser

2025

Developed an adaptive system that observes SQL query workloads and incrementally optimises physical data layouts (partitioning and sorting) for databases. Implemented workload logging, layout proposal, dataset rewriting, and performance evaluation in a closed feedback loop, allowing the system to learn which layouts perform best under changing access patterns.

QueryGPT: NL-to-SQL Pipeline

2025

Built a multi-agent natural language to SQL pipeline that decomposes user queries into intent detection, table selection, column pruning, and query generation. Focused on reducing token usage and failure modes by enforcing structured intermediate outputs. Integrated 'llama-index', 'pydantic-ai', and a Neo4j vector store to support retrieval-augmented generation and improve maintainability as schemas evolve.

Markov Decision Process Framework

2025

Implemented a framework for defining and solving finite-horizon Markov Decision Processes with time-dependent state representations for revenue optimisation. Designed utilities for estimating stochastic transition matrices from observational data (Poisson, Normal, Exponential distributions), computing optimal pricing policies via backward induction, and generating revenue forecasts through forward simulation under capacity constraints.

Languages & tools

Proficient: Python, SQL, GBT, Polars, PyTorch, EconML, statsmodels, AWS, Docker, Git, Linux **Experienced:** PySpark, Rust, C++

Additional Experience

- Scrum Master Certificate 2025